

Layout Draft

Design Note

EBeam_weijiany | five course foundation designs
Reference waveguide, three MZIs, and one single-bus ring

Objective. Measure the TE waveguide group index from measured free spectral range (FSR), compare three actual MZI path differences, and use a two-GC reference to assess the shared coupling envelope. The ring provides a complementary resonance-based measurement.

Design basis

ITEM	ACTUAL LAYOUT / DESIGN TARGET
Process and waveguide	220 nm SOI; single full etch; oxide cladding; Strip TE 1550 nm, w=500 nm. Modal behavior is unverified.
Footprint	605 x 410 um; Si mask layer 1/0; top cell EBeam_weijiany.
Components	5 circuits and 10 GCs; installed EBeam PDK 0.4.53.
Test geometry	127 um vertical GC pitch; lower GC input, upper GC output; GC waveguide ports face right.

Native routed-path parameter set

DESIGN	ACTUAL PATH LENGTHS (um)	FSR ESTIMATE (nm)
REF	Reference routed length 163.897	No intended fringe period
MZI15	L1 125.797; L2 156.797 Delta L 31.000	17.22 - 19.38
MZI25	L1 125.797; L2 176.797 Delta L 51.000	10.47 - 11.78
MZI35	L1 125.797; L2 196.797 Delta L 71.000	7.52 - 8.46
RING10	Loop 68.313 R 10.0; gap 200 nm	7.82 - 8.79

The course planning objective is FSR < 50 nm; the largest estimate here is 19.38 nm, conditional on the assumed group index. FSR intervals use **assumed** n_g = 4.0 - 4.5 at 1550 nm. They are planning estimates, not simulated or measured spectra. MZI names identify routing settings; the table gives the real PCell curve-length differences. Ring loop length includes the actual closure route and is not assumed to be 2*pi*R.

Layout and measurement labels

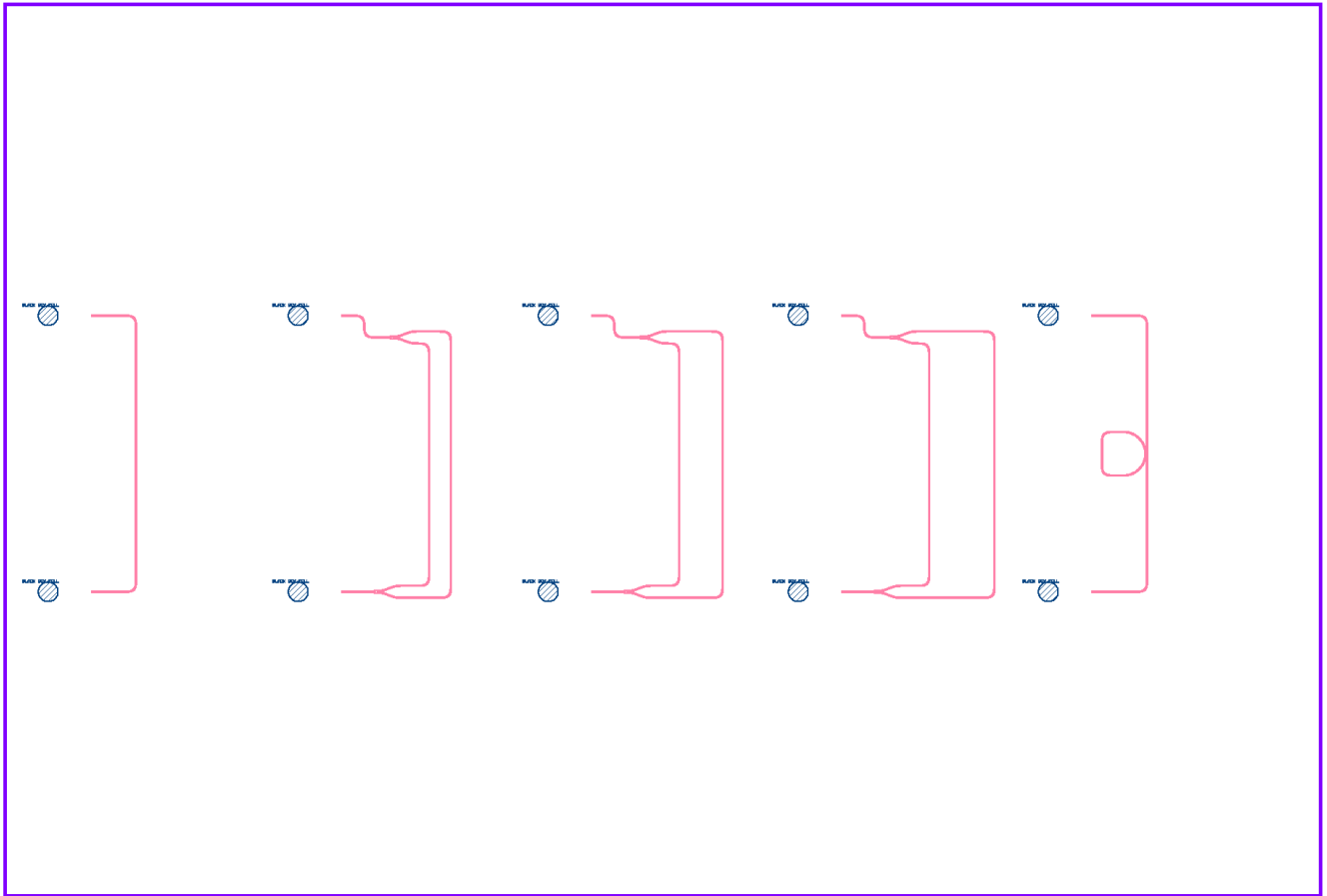


Figure 1. Clean native KLayout preview of the actual draft: silicon, floorplan, official black-box outlines, and fiber targets. Text and device/pin-recognition annotations are hidden in this image; the original GC black-box placeholders and measurement labels remain in the GDS. Left to right: REF, three MZIs, RING10. Geometry is in μm ; this is a layout view, not an optical field.

DESIGN	GC X (μm)	INPUT TEST LABEL
REF	40	opt_in_TE_1550_device_weijiany_REF
MZI15	155	opt_in_TE_1550_device_weijiany_MZI15
MZI25	270	opt_in_TE_1550_device_weijiany_MZI25
MZI35	385	opt_in_TE_1550_device_weijiany_MZI35
RING10	500	opt_in_TE_1550_device_weijiany_RING10

Fiber-array mapping

Each pair places the input at $y = 140 \mu\text{m}$ and the output at $y = 267 \mu\text{m}$. The input is the second GC from the top, consistent with course channel 2 input and channel 1 output. Each input has one unique `opt_in_TE_1550_device_weijiany_COMMENT` label at the GC origin. No third output or alternate test fixture is assumed.

PDK hierarchy

`GC_TE_1550_8degOxide_BB` and `ebeam_y_1550` retain their original library names and local bounding boxes in exported files. MZI topology follows the installed official MZI example; the ring uses the official half-ring component and `ring_singlebus` topology.

Validation of this draft

CHECK	EVIDENCE AND RESULT
Generator	PASSED; KLayout 0.30.12; SiEPIC 0.5.31; EBeam 0.4.53.
Functional / DFT	PASSED: 0 reported errors; native SiEPIC layout_check with verify_DFT=True.
Native physical DRC	PASS: 0 native report items; installed official EBeam rule deck.
Silicon boundary	Outside floorplan: 0.000000 um2; silicon area: 805.79 um2.
Export readback	PASS; one named top cell, unchanged DBU, zero silicon symmetric difference, preserved official BB names and local bboxes.
File identity	The PDF builder independently rehashed every listed export and the supplied DRC input/report.
Optical simulation	not run; no mode or circuit result is claimed.

The installed native rule deck uses an effective silicon width/space check of 69 nm. Its original rules are retained and no black-box exclusions are added. This is distinct from the course checklist's stated 60 nm minimum.

Deliverable identity

FILE	BYTES	SHA-256 PREFIX (16 HEX)
EBeam_weijiany_editable.gds	98,958	81132e1e3795bafa
EBeam_weijiany.gds	94,318	10f733facbfa1056
EBeam_weijiany.oas	19,861	2fe3b82f98dc6e1d

Scope. These checks assess generated geometry, installed-deck DRC, and SiEPIC functional/DFT rules only. They do not establish modal confinement, optical performance, current course-batch acceptance, final fabrication approval, or successful GitHub/edX submission.

Model and experimental plan

Ideal MZI model

For ideal equal-amplitude arms and a fixed splitter/combiner phase, the normalized transmission is:

$$T(\lambda) = [1 + \cos(\Delta\phi)] / 2$$

$$\Delta\phi = 2\pi n_{\text{eff}}(\lambda) \Delta L / \lambda$$

$$n_g(\lambda) = n_{\text{eff}} - \lambda (dn_{\text{eff}} / d\lambda)$$

$$n_g \sim \lambda^2 / (FSR * |\Delta L|)$$

Use consistent units and the measured center wavelength. The last expression is a local-FSR approximation. Bends, splitter phase dispersion, loss imbalance, and different straight/bend group indices may make the extracted value an effective path average. For the ring, substitute its actual loop length for $|\Delta L|$ and account for coupler phase dispersion.

Sweep and de-embedding procedure

1. Measure the two-GC reference and all device transmissions with the same TE alignment, polarization, fiber-array channel, power convention, and wavelength grid; retain raw spectra.
2. Divide device power by the reference power in linear units over the common reliable band. This estimates the shared GC envelope; route-length mismatch and device-dependent losses remain.
3. Identify several adjacent MZI peaks or valleys, compute local FSR near 1550 nm, and use the actual ΔL values on page 1. Compare the three estimates and propagate wavelength/FSR uncertainty. Reject intervals without sufficient reliable fringes.
4. Extract the ring's adjacent resonance spacing using the actual loop length. Compare its effective n_g with the MZI estimates; do not infer coupling strength, loss, or Q from layout alone.

Remaining before a complete design proposal

Actual TE mode profiles and n_{eff}/n_g dispersion; a circuit solver using characterized PDK components; bend/coupler loss and phase; process-variability evaluation; spectra and uncertainty. This note accompanies the layout draft and does not replace those results.

Sources and traceability

[Course layout submission instructions](#) | [Course space, testing, and checklist](#) | [Official EBeam PDK v0.4.53](#) | [Official SiEPIC-Tools](#) | [Course openEBL repository](#)

Source records: layout-metadata.json, native functional RDB, and the supplied official DRC summary/provenance. Full SHA-256 values and native curve-length sources are retained in the JSON records; this PDF shows shortened hashes for readability.